Lesson Planning

Complete Lesson Plan Book

Class: Class 10

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Learning Objectives

- Students will identify the coefficients a, b and c in any quadratic equation.
- Students will state and apply the quadratic formula correctly.
- Students will solve quadratic equations that do not factorise easily.
- Students will use the discriminant to predict the nature of the roots.

Key Concepts

- A quadratic equation has the standard form $ax^2 + bx + c = 0$, where a is never zero.
- The quadratic formula $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$ gives the roots of any quadratic equation.
- The part $b^2 - 4ac$ is called the discriminant: positive means two real roots, zero means one repeated root, negative means no real roots.
- The formula works even when the equation cannot be factorised into simple brackets.

Teaching Methodology (Step by Step)

1. Recall: solve $x^2 - 5x + 6 = 0$ by factorisation on the board, then ask what to do when factorisation is hard.
2. Write the standard form $ax^2 + bx + c = 0$ and identify a, b, c in $2x^2 + 3x - 5 = 0$ together with the class.
3. Present the quadratic formula step by step and explain each symbol, especially the ± sign.
4. Worked example: solve $x^2 + 4x + 2 = 0$ fully, showing substitution, simplification and both roots.
5. Introduce the discriminant with three quick examples showing two roots, one root and no real roots.
6. Guided practice: students solve $x^2 - 3x - 4 = 0$ while the teacher circulates and corrects.

Classroom Activities

- Formula race: pairs compete to correctly identify a, b, c from five flashed equations.
- Board relay: each team member completes one step of solving an equation with the formula.
- Discriminant detective: given ten equations, students sort them by the nature of their roots without solving fully.

Assessment Questions

1. Identify a, b and c in $3x^2 - 7x + 2 = 0$.
2. Use the quadratic formula to solve $x^2 + 6x + 5 = 0$.
3. Find the discriminant of $2x^2 + 4x + 2 = 0$ and state the nature of its roots.
4. Why is the quadratic formula useful when factorisation fails?

Homework

- Solve by the quadratic formula: $x^2 - 5x + 3 = 0$ and $2x^2 + x - 6 = 0$.
- For $3x^2 + 2x + 5 = 0$, compute the discriminant and explain what it tells you.

Time Allocation

40 minutes